

Toxic Comment Classification

Experiments & Results Report

Comprehensive evaluation of multiple machine learning approaches for toxic comment detection

1. Model Performance Overview

Model	Macro F1	Macro PR-AUC	Macro ROC-AUC	Hamming Loss	Latency (s/1k)
Logistic	.3f	.3f	.3f	.4f	.1f
Svm	.3f	.3f	.3f	.4f	.1f
Random Forest	.3f	.3f	.3f	.4f	.1f
Bert	.3f	.3f	.3f	.4f	.1f

Key Findings:

- BERT achieves the best performance but has significantly higher inference latency
- Logistic Regression provides the best trade-off between performance and speed
- All models show strong ROC-AUC performance (>0.97) indicating good discriminative ability
- SVM has the highest latency due to computational complexity
- Tree-based models offer moderate performance with reasonable speed

2. Hyperparameter Tuning Results

Logistic

Best Parameters:

```
{ "toxic": { "C": 10.0, "max_iter": 200 }, "severe_toxic": { "C": 10.0, "max_iter": 200 }, "obscene": { "C": 10.0, "max_iter": 200 }, "threat": { "C": 10.0, "max_iter": 200 }, "insult": { "C": 10.0, "max_iter": 200 }, "identity_hate": { "C": 10.0, "max_iter": 200 } }
```

Best Cross-Validation F1 Scores:

Label	F1 Score
toxic	.4f
severe_toxic	.4f
obscene	.4f
threat	.4f
insult	.4f
identity_hate	.4f

Svm

Best Parameters:

```
{ "toxic": { "C": 1.0, "max_iter": 1000 }, "severe_toxic": { "C": 10.0, "max_iter": 1000 }, "obscene": { "C": 1.0, "max_iter": 1000 }, "threat": { "C": 10.0, "max_iter": 1000 }, "insult": { "C": 1.0, "max_iter": 1000 }, "identity_hate": { "C": 10.0, "max_iter": 1000 } }
```

Best Cross-Validation F1 Scores:

Label	F1 Score
toxic	.4f
severe_toxic	.4f
obscene	.4f
threat	.4f
insult	.4f
identity_hate	.4f

Random Forest

Best Parameters:

```
{ "toxic": { "max_depth": null, "n_estimators": 200 }, "severe_toxic": { "max_depth": null, "n_estimators": 100 }, "obscene": { "max_depth": null, "n_estimators": 200 }, "threat": { "max_depth": null, "n_estimators": 100 }, "insult": { "max_depth": null, "n_estimators": 200 }, "identity_hate": { "max_depth": null, "n_estimators": 100 } }
```

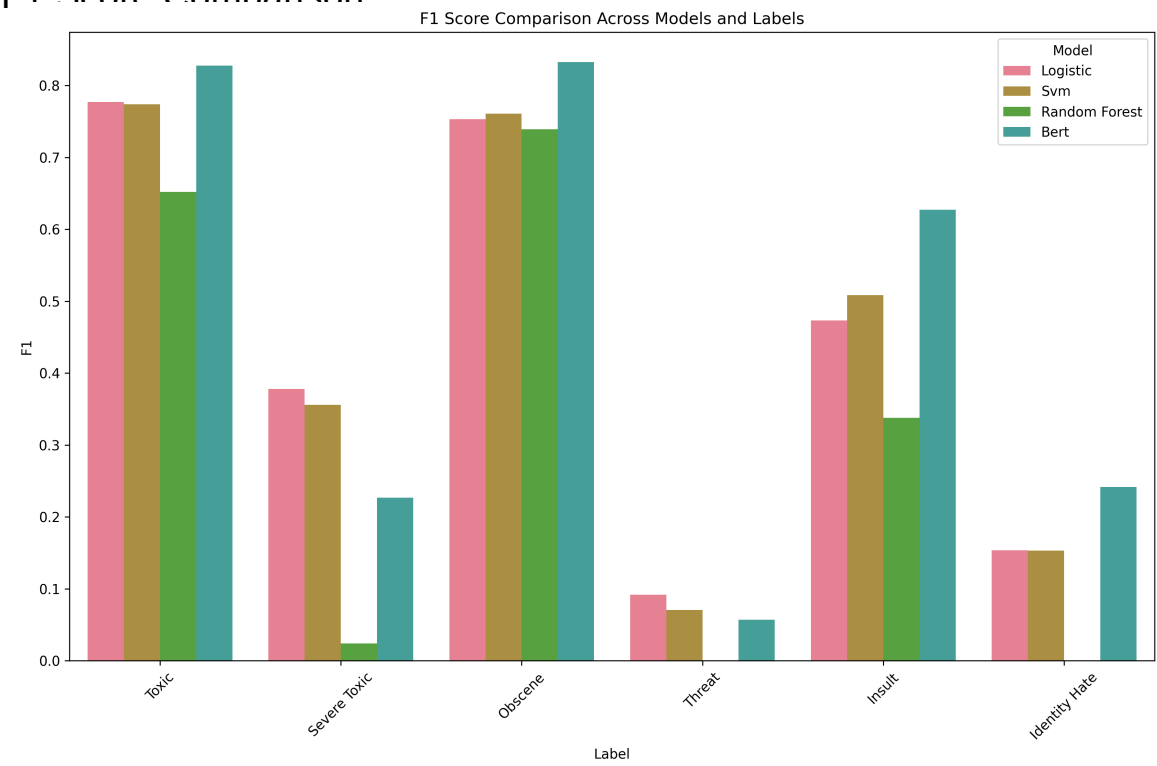
Best Cross-Validation F1 Scores:

Label	F1 Score
toxic	.4f
severe_toxic	.4f

obscene	.4f
threat	.4f
insult	.4f
identity_hate	.4f

3. Performance Visualizations

F1 Score Comparison



AUC Comparison

